

Dietary fiber intake and risk of breast cancer by menopausal and estrogen receptor status

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Purpose Evaluate the hypothesis that relation of breast cancer associated with dietary fiber intakes varies by type of fiber, menopausal, and the tumor's hormone receptor status. **Methods** A case-control study of female breast cancer was conducted in Connecticut. A total of 557 incident breast cancer cases and 536 age frequency-matched controls were included in the analysis. Information on dietary intakes was collected through in-person interviews with a semi-quantitative food frequency questionnaire and was converted into nutrient intakes. Odds ratios and 95% confidence intervals were estimated by unconditional logistic regression. **Results** Among pre-menopausal women, higher intake of soluble fiber (highest versus lowest quartile of intake) was associated with a significantly reduced risk of breast cancer (OR = 0.38, 95% CI, 0.15–0.97, Ptrend = 0.08). When further restricted to pre-menopausal women with ER– tumors, the adjusted OR for the highest quartile of intake was 0.15 (95% CI, 0.03–0.69, Ptrend = 0.02) for soluble fiber intake. Among post-menopausal women, no reduced risk of breast cancer was observed for either soluble or insoluble fiber intakes or among ER+ or ER– tumor groups. **Conclusions** The results from this study show that dietary soluble fiber intake is associated with a significantly reduced risk of ER– breast cancer among pre-menopausal women. Additional studies with larger sample size are needed to confirm these results.